Student Performance Analytics System

EWB Courses | Python with AI

1. Project Title

Item	Details
Project Title	Student Performance Analytics System
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College / Department	Info Institute of Engineering – B.Tech Information Technology
Course	Python with AI – EWB Courses
Submission Date	12th Oct 2026

2. Objective

Calculating student marks, grades and results manually is slow and easy to get wrong. The objective of this project is to build a Python program that reads student data from a file and automatically produces useful results: total and average marks, grades, pass/fail status, subject-wise performance, department-wise performance and the top-performing students.

The project solves the problem of quickly understanding how a class is performing, which students need attention, and which subjects are strong or weak.

3. Technologies Used

- **Python 3** – core programming language (functions, loops, conditions)
- **NumPy** – numerical calculations such as sum, mean, max, min and comparisons
- **Pandas** – loading the CSV, DataFrame operations, sorting, filtering and groupby
- **VS Code** – code editor used to write and run the program
- **Git & GitHub** – version control and project submission

4. Dataset Description

The dataset is stored in students.csv and contains 25 student records with no missing values.

Column	Description
Student_ID	Unique ID of the student (S001 – S025)
Name	Name of the student
Department	CSE, IT, ECE, EEE or MECH
Maths, Science, English, Computer, Tamil	Marks out of 100 in each of the five subjects
Attendance	Attendance percentage

The program adds four calculated columns: **Total, Average, Grade** and **Result**. The processed data is saved to student_results.csv.

5. Implementation

1. **Data loading:** The CSV file is read with pd.read_csv(). The columns, number of records and missing values are displayed to check the data.
2. **Total and average:** The five subject columns are converted to a NumPy array. np.sum(axis=1) gives each student's total and np.mean(axis=1) gives the average.
3. **Grading:** The function get_grade() assigns a grade using the average marks (rule below).
4. **Pass/Fail:** The function get_status() uses np.all() – a student passes only if the marks are 40 or above in every subject.
5. **Loops:** A for loop goes through each student to assign Grade and Result. Another for loop goes through each subject to calculate highest, lowest, average and pass count.
6. **Analysis:** Pandas functions (idxmax, idxmin, nlargest, value_counts, groupby) answer the questions: who has the highest average, which subject is best, how many passed, and which department performs best.
7. **Output:** Results are printed with clear headings so they can be understood without reading the code.

Grading rule (based on average marks):

Average Marks	Grade	Remark
90 and above	A+	Outstanding
80 – 89.99	A	Excellent
70 – 79.99	B	Good
60 – 69.99	C	Satisfactory
50 – 59.99	D	Needs improvement
Below 50	F	Fail grade

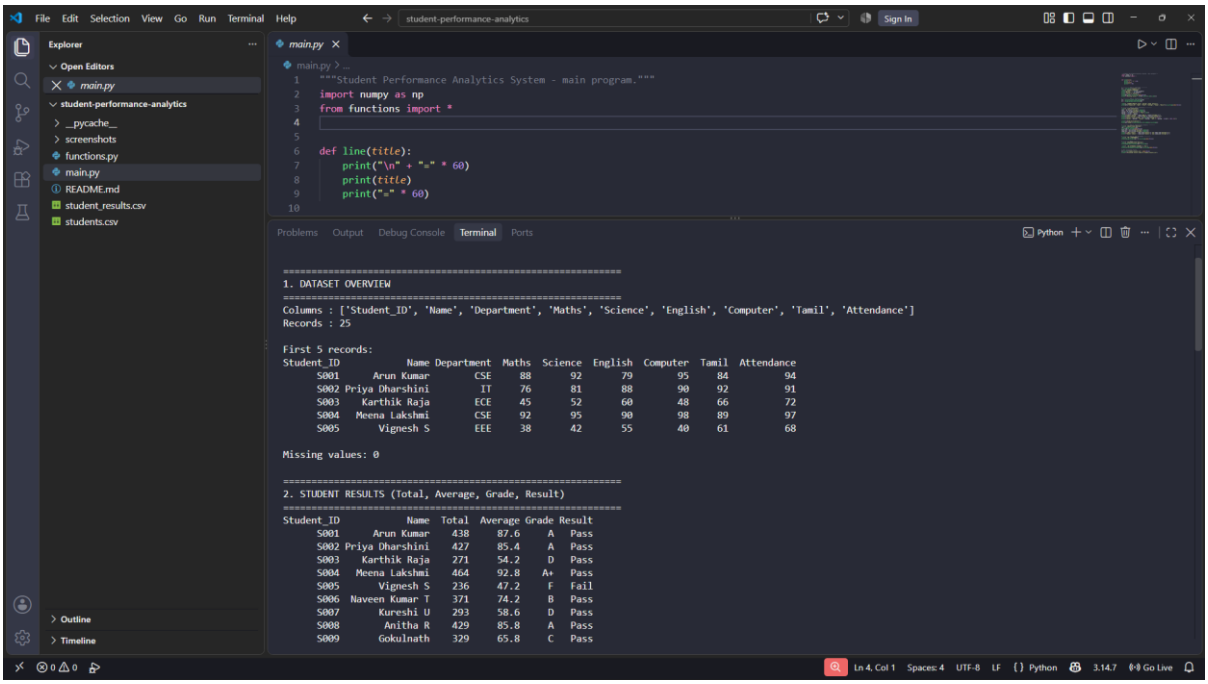
Pass condition: A student is marked Pass only if the marks are at least 40 in all five subjects; otherwise Fail.

6. Key Features

Function	Purpose
load_data()	Reads the student CSV file into a DataFrame
calculate_total_and_average()	Calculates total and average marks using NumPy
get_grade()	Applies the grading rule
get_status()	Applies the pass/fail condition
assign_grades_and_status()	Loops through students to assign grade and result
subject_analysis()	Highest, lowest, average and pass count for each subject
top_performers()	Finds the top N students by average
department_analysis()	Department-wise average and pass count
low_attendance_students()	Lists students with attendance below 75%

7. Output / Screenshots

The program is run in VS Code using the command python main.py. The screenshots below show the complete output.



```
"""Student Performance Analytics System - main program."""
import numpy as np
from functions import *

def line(title):
    print("\n" + "=" * 60)
    print(title)
    print("=" * 60)
```

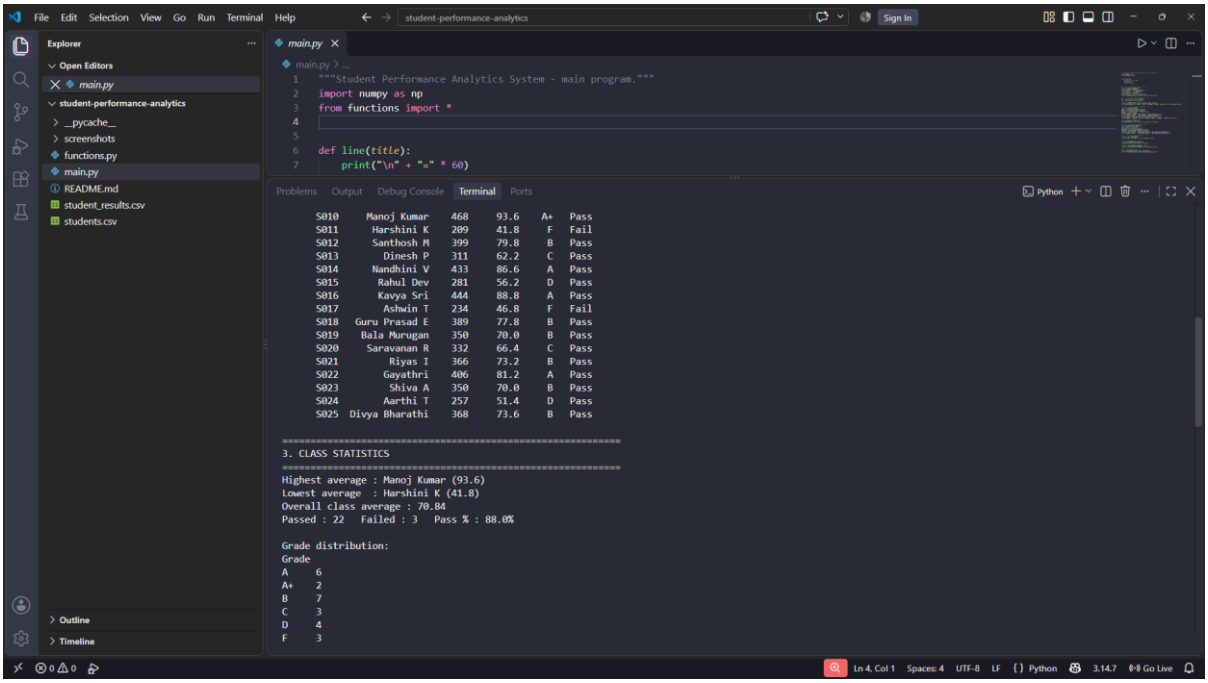
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1. DATASET OVERVIEW
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Columns : ['Student_ID', 'Name', 'Department', 'Maths', 'Science', 'English', 'Computer', 'Tamil', 'Attendance']
Records : 25

First 5 records:
Student_ID           Name Department  Maths  Science  English  Computer  Tamil  Attendance
      S001     Arun Kumar        CSE     88       92       79        95     84          94
      S002 Priya Dharshini         IT     76       81       88        90     92          91
      S003    Karthik Raja        ECE     45       52       60        48     66          72
      S004   Meena Lakshmi        CSE     92       95       90        98     89          97
      S005       Vignesh S        EEE     38       42       55        40     61          68

Missing values: 0

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2. STUDENT RESULTS (Total, Average, Grade, Result)
============================================================
Student_ID           Name  Total  Average Grade Result
      S001     Arun Kumar    438     87.6     A   Pass
      S002 Priya Dharshini    427     85.4     A   Pass
      S003    Karthik Raja    271     54.2     D   Pass
      S004   Meena Lakshmi    464     92.8    A+   Pass
      S005       Vignesh S    236     47.2     F   Fail
      S006  Naveen Kumar T    371     74.2     B   Pass
      S007       Kureshi U    293     58.6     D   Pass
      S008        Anitha R    429     85.8     A   Pass
      S009       Gokulnath    329     65.8     C   Pass
```

Figure 1: Dataset overview and student results (first part)



```
      S010     Manoj Kumar    468     93.6    A+   Pass
      S011      Harshini K    209     41.8     F   Fail
      S012      Santhosh M    399     79.8     B   Pass
      S013        Dinesh P    311     62.2     C   Pass
      S014      Nandhini V    433     86.6     A   Pass
      S015       Rahul Dev    281     56.2     D   Pass
      S016       Kavya Sri    444     88.8     A   Pass
      S017       Ashwin T     234     46.8     F   Fail
      S018   Guru Prasad E    389     77.8     B   Pass
      S019    Bala Murugan    350     70.0     B   Pass
      S020     Saravanan R    332     66.4     C   Pass
      S021         Riyas I    366     73.2     B   Pass
      S022        Gayathri    406     81.2     A   Pass
      S023         Shiva A    350     70.0     B   Pass
      S024        Aarthi T    257     51.4     D   Pass
      S025  Divya Bharathi    368     73.6     B   Pass

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3. CLASS STATISTICS
============================================================
Highest average : Manoj Kumar (93.6)
Lowest average  : Harshini K (41.8)
Overall class average : 70.84
Passed : 22   Failed : 3   Pass % : 88.0%

Grade distribution:
Grade
A     6
A+    2
B     7
C     3
D     4
F     3
```

Figure 2: Student results (remaining records), class statistics and grade distribution

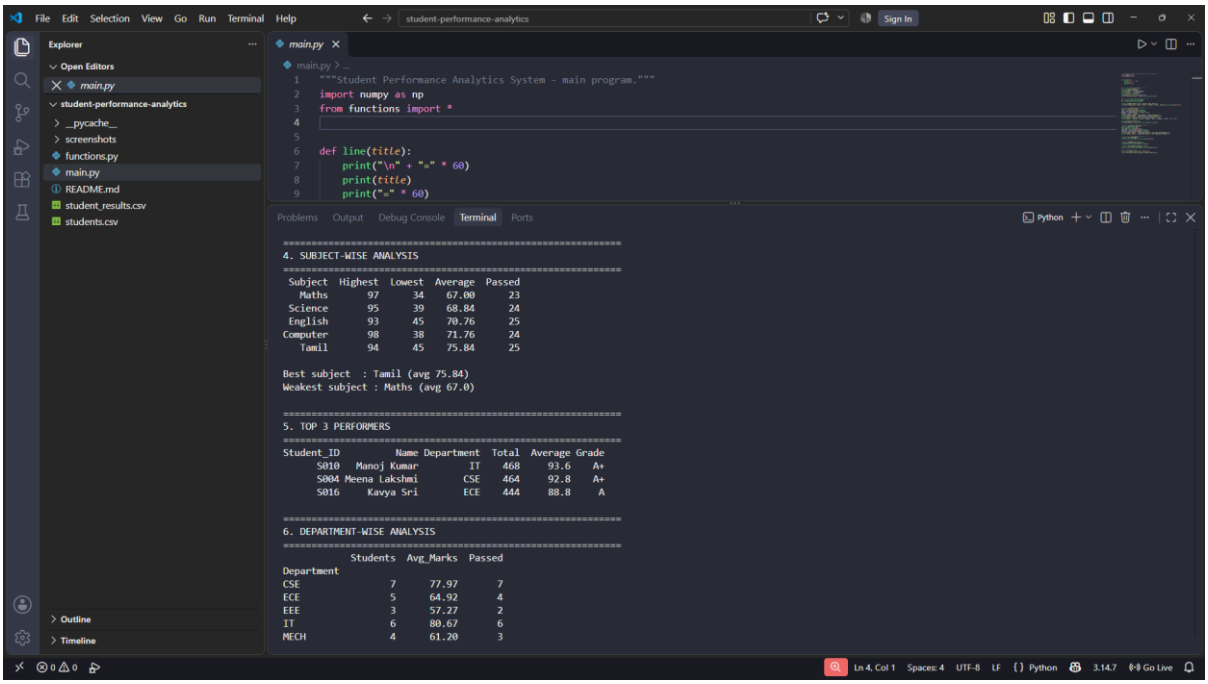


Figure 3: Subject-wise analysis, top 3 performers and department-wise analysis

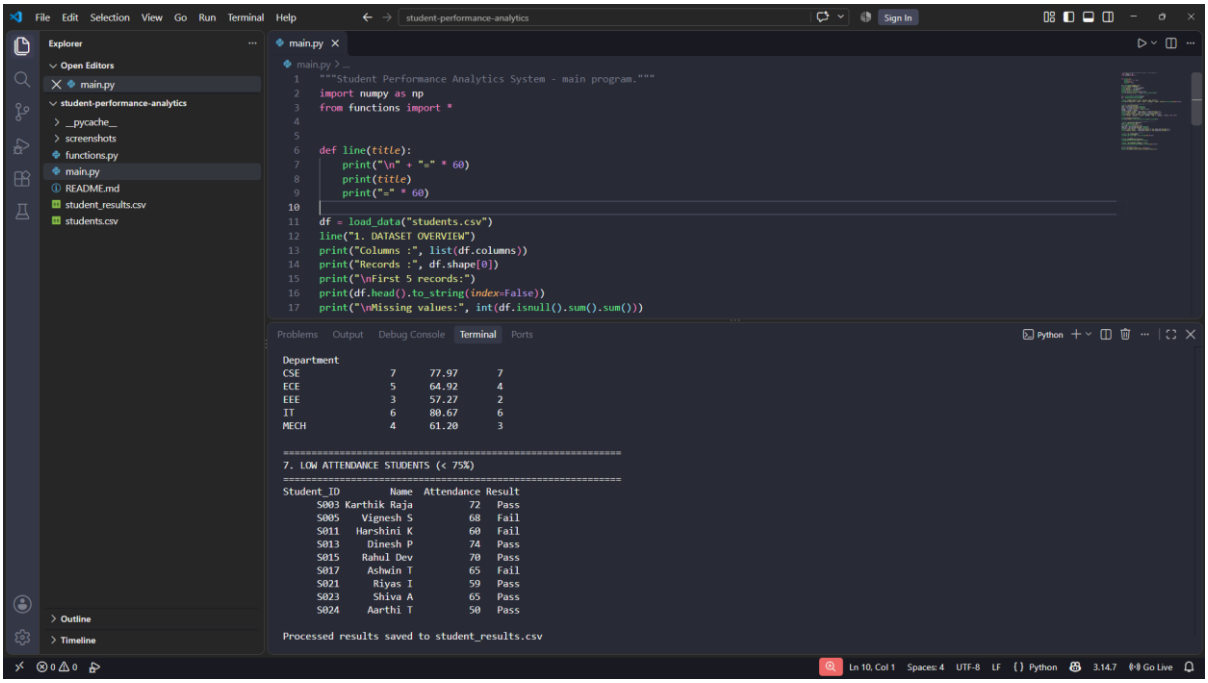


Figure 4: Department-wise analysis and low attendance students

8. Final Outcome

The project successfully analysed the performance of 25 students and produced the following results:

- Highest average: Manoj Kumar (93.6); lowest average: Harshini K (41.8)
- Overall class average: 70.84
- 22 students passed and 3 failed (pass percentage 88%)
- Best subject: Tamil (average 75.84); weakest subject: Maths (average 67.00)
- Top 3 performers: Manoj Kumar, Meena Lakshmi and Kavya Sri

- Best department by average marks: IT (80.67); lowest: EEE (57.27)
- 9 students have attendance below 75%

The program runs without errors and presents all results in a clear, readable format.

9. Challenges & Learning

Challenge	Solution / Learning
Deciding a clear pass condition	Defined that a student must score at least 40 in every subject, and implemented it with np.all().
Choosing between NumPy and Pandas	Used NumPy for calculations (sum, mean, max, min) and Pandas for loading, sorting, filtering and groupby.
Output too long to read	Divided the output into numbered sections with headings and separator lines.
Keeping code organised	Moved reusable logic to functions.py and kept main.py only for running and displaying results.

What I learned: writing reusable functions, using loops with DataFrames, applying NumPy for numerical work, analysing data with Pandas, and publishing a project on GitHub with a README.

10. GitHub Repository

Link : https://github.com/KasimBasi05/Student-Performance-Analyzer